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Synthetic Influenza vaccine (FLU-v) stimulates cell mediated immunity in a double-blind, randomised, placebo-controlled Phase I trial☆

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ABSTRACT

Objectives: Current Influenza vaccines elicit antibody mediated prophylactic immunity targeted to viral capsid antigens. Despite their global use these vaccines must be administered yearly to the population, cannot be manufactured until the circulating viral strain(s) have been identified and have limited efficacy. A need remains for Influenza vaccines addressing these issues and here we report the results of a Phase Ib trial of a novel synthetic Influenza vaccine (FLU-v) targeting T cell responses to NP, M1 and M2.
Methods: Forty-eight healthy males aged 18–40 were recruited for this single-centre, randomised, double blind study. Volunteers received one single low (250 μg) or high (500 μg) dose of FLU-v, either alone or adjuvanted. Safety, tolerability and basic immunogenicity (IgG and IFN-γ responses) parameters were assessed pre-vaccination and for 21 days post-vaccination.
Results: FLU-v was found to be safe and well tolerated with no vaccine associated severe adverse events. Dose-dependent IFN-γ responses >2-fold the pre-vaccination level were detected in 80% and 100% of volunteers receiving, respectively, the low and high dose adjuvanted FLU-v formulations. No formulation tested induced any significant FLU-v antibody response.
Conclusion: FLU-v is safe and induces a vaccine-specific cellular immunity. Cellular immune responses are historically known to control and mitigate infection and illness during natural infection.
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1. Introduction

Influenza causes yearly epidemics associated with significant morbidity and economic loss [1] as well as increased mortality in some groups (e.g. elderly) [2]. Occasionally, it also causes pandemics characterised by high infection and mortality rates. The Spanish Flu pandemic of 1918 affected one-third of the world population (i.e. 500 million) and caused over 40 million fatalities [3].

For over 50 years, Influenza public health programs have relied on vaccines that elicit neutralising antibodies to Haemagglutinin (HA) and Neuraminidase (NA). These prophylactic vaccines have, however, three major shortcomings. Firstly, the high variability of HA and NA means that one vaccine cannot provide universal protection against all Influenza strains. As new variants of the virus emerge every year, vaccine preparations must be updated and populations re-vaccinated. Secondly, mass manufacturing of HA/NA-based Influenza vaccines can only start once the circulating pathogenic strain(s) is identified. This results in a delay of 6 months or more between an Influenza outbreak and a vaccine being available [4]. Finally, the efficacy of HA/NA-based vaccines is limited. A recent report [5] suggests their efficacy rate is only 59% in 18–64 year olds, falling to 35% in those over 65 years of age.

Much research has been carried out on novel Influenza vaccine candidates that totally or partially address the above described shortcomings of current vaccines (reviewed in [6] and [7]). Most of them continue to target antibody immunity to viral antigens (e.g. HA, NA, M2e), whilst a few target cellular immune responses to a different range of antigens (e.g. NP, M1, M2).

Influenza-specific $CD8^+$ T-cell responses are a feature on natural infection [8] and we and others [9–13] have shown that a $CD8^+$ T-cell response can, in the absence of neutralising antibodies, protect mice against lethal challenge with homotypic and heterotypic Influenza strains. In our case [9], this protective response was induced following immunisation with an equimolar admixture of synthetic polypeptides covering conserved immunogenic regions in M1, M2, NP and PB1. We now describe the results of a first-in-man Phase I clinical trial of a novel Influenza candidate, FLU-v, based on those synthetic polypeptides.

2. Materials and methods

2.1. Study population

FLU-v was assessed in a randomised, double blind single-centre study (Quintiles Drug Research Unit at Guy's Hospital).

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Volunteers were enrolled if at screening they were males aged 18–40 years with Body Mass Index (BMI) 18.5–28.5 kg/m^2, no clinically significant abnormal findings (i.e. physical examination, electrocardiogram (ECG), medical history or laboratory results) and non-users or low-users (i.e. ≤10 cigarettes per day) of tobacco-products.

2.2. Vaccine description

FLU-v is a sterile equimolar mixture of four synthetic polypeptides:

M1 (32 aa) DLEALMEWLKTRPILSPLTKGILGFVFTLTVP
NPA (21 aa) DLIFLARSALILRGSVAHKSC
NPB (20 aa) PGIADIEDLTLLARSMVVVR
M2 (24 aa) IIGILHLILWILDRLFFKCIYRLF

These sequences represent potentially immunoreactive conserved regions identified in silico through multiple sequence (ClustalW) [14] and immunogenicity analysis [9] of all Influenza A and B protein sequences available at the National Centre for Biotechnology Information (NCBI) taxonomy database (December 2005). Each polypeptide represents a region of the consensus sequence where every consecutive amino acid is present in ≥70% of the sequence population tested and where >5 potential human T cell epitopes are found.

Polypeptides were synthetically manufactured (Bachem AG, Bübendorf, Switzerland) in accordance with current Good Manufacturing Practice.

The vaccine was delivered in 1.0 ml volume as a single Low (250 μg) or High (500 μg) dose either in saline or saline emulsified with adjuvant ISA-51 (Seppic, France). The Placebo was either saline alone or ISA-51 emulsified saline.

2.3. Study procedures

At drug administration, forty-eight eligible volunteers were randomised sequentially into one of the following six treatment groups (ratio 5:5:5:5:2:2): FLU-v 250 μg (LD), FLU-v 250 μg in adjuvant (LDA), FLU-v 500 μg (HD), HIV-v 500 μg in adjuvant (HDA), saline (Placebo-S) and adjuvanted saline (Placebo-SA).

Participants were immunised subcutaneously on Day 1 and monitored for 21 days. Clinical laboratory tests (clinical chemistry, haematology and urinalysis) were performed at screening and on Days −1, 1, 2, 14 and 21. Subjects were released on Day 1 following completion of the laboratory assessments. ECGs, vital signs and local tolerability were assessed at screening and pre-dose (Day 1) and post-dose on Days 1 and 2. Volunteer self-recorded daily observations were assessed by the clinical staff on Day 7 and any abnormalities were entered into the AE database.

2.4. Regulatory approval and ethical considerations

The study was conducted in accordance with Good Clinical Practice, the Declaration of Helsinki (1964 and 2008), and applicable regulatory requirements. Trial registered under ClinicalTrials.gov Identifier NCT01181336. Written informed consent was obtained from all participants.

2.5. Study objectives and endpoints

The objectives of the study were to determine the safety, tolerability and basic immunogenicity (antibody and cellular responses) of two dose levels of FLU-v, formulated either alone or in adjuvant.

IgG responses were determined as the Mean OD_{450} of the sera samples from a specified time point (pre-vaccination and Days 14 and 21 post-vaccination) diluted 1/200 in PBS and tested (in triplicate for pre-vaccination samples and in duplicate for post-vaccination samples) in a validated direct ELISA based on a protocol described elsewhere [9]. Volunteers whose post-vaccination Mean OD_{450} was above the cut-point were considered to have developed FLU-v specific antibody immunity. Each volunteer's cut-point was determined as 2× Mean OD_{450} pre-vaccination.

IFN-g responses of thawed PBMCs were determined by validated IFN-γ ELISA (BD, Human IFN-g kit 555142) following 24 h in vitro stimulation with FLU-v. PBMCs were isolated on day on blood collection and frozen. Response levels were calculated in triplicate as pg/ml of IFN-γ produced against a concentration standard. Any individual whose post-vaccination Flu-v-specific response was ≥2-fold higher than his pre-vaccination response was considered to have developed a vaccine specific cellular immune response.

2.6. Statistical analysis

Inter-group differences were determined using the Mann–Whitney test. Forty-eight subjects were considered appropriate to determine safety and tolerability. No formal sample size calculations were made as no previous data is available to estimate expected changes for this treatment.

2.7. Safety reviews and interim analysis

Reviews to evaluate safety and tolerability and allow continuation of the study were completed by the Safety Committee (independent clinicians) 72 h after one volunteer from each experimental group was vaccinated.

3. Results

3.1. Recruitment and safety

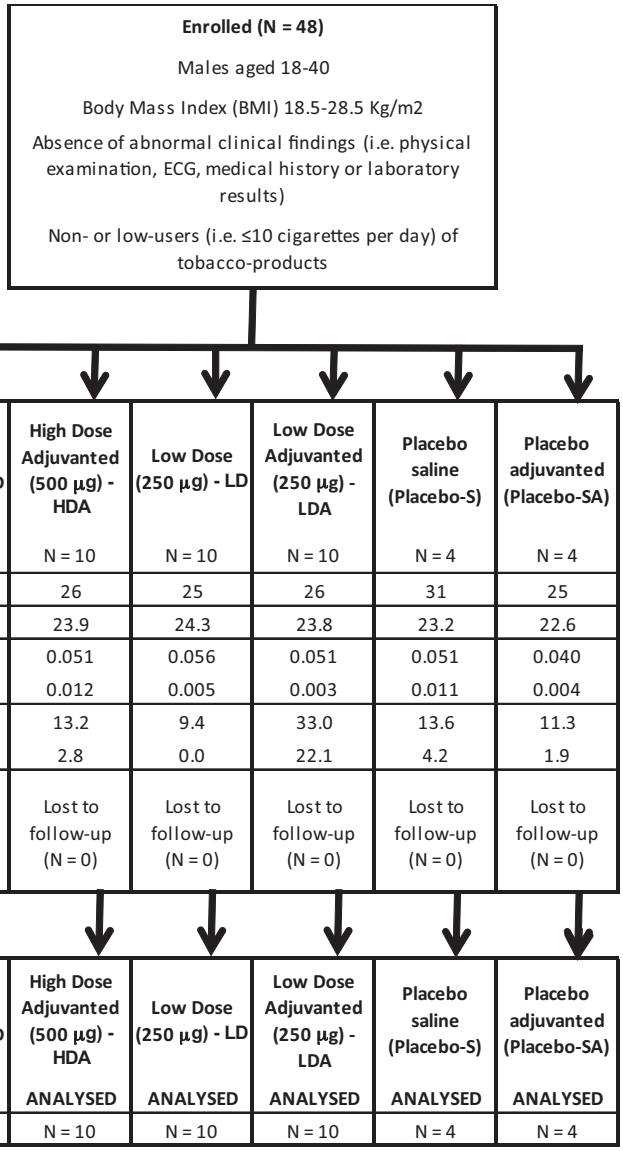
Forty-eight volunteers were enrolled in this study (April'10 to July'10). Trial profile and baseline clinical and demographic data are reported in Fig. 1.

For the non-adjuvanted FLU-v formulations, nine individuals (90%) receiving the low dose (LD group) reported 24 AEs, whilst eight individuals (80%) receiving the high dose (HD group) reported 38 AEs. For the adjuvanted FLU-v formulations, nine individuals (90%) receiving the low dose (LDA group) reported 25 AEs, whilst 10 individuals (100%) receiving the high dose (HDA group) reported 42 AEs. In the placebo groups, all eight individuals (100%) reported AEs with 8 and 12 cases, respectively, for the saline (Placebo-S) and saline in adjuvant (Placebo-SA) groups.

In total 149 adverse events (AEs) were reported, of which 86% (128/149) were MILD, 13% (19/149) were MODERATE and 1% (2/149) were SEVERE. Pain at the site of injection was the most common MILD and MODERATE AE with 43.4% (56/129) and 52.4% (11/19) of cases, respectively. Neither of the two SEVERE AEs reported (i.e. appendicitis and pre-syncope) was considered to be directly caused by the vaccine.

The majority of AEs were reported in the system organ class of general disorders and administration site conditions. The frequency of AEs within this category was 63% (15/24) for the LD group, 76% (29/38) for the LDA group, 60% (15/25) for the HD group, 81% (34/42) for the HDA group, 75% (6/8) for the Placebo-S group and 58% (7/12) for the Placebo-SA group.

Headache, malaise, fatigue, myalgia, nausea, vomiting, diarrhoea, chills, fever and other (as reported by the subject) were assessed for systemic tolerability. The most common AEs reported in this area were headache [LD: 17% (4/24), LDA: 5% (2/38), HD: 12% (3/25), HDA: 5% (2/42), Placebo-S: 12% (1/8) and Placebo-SA: 8% (1/12)], fatigue [LD: 12% (3/24), LDA: 8% (3/38), HD: 8% (2/25),



		High Dose (500 µg) - HD	High Dose Adjuvanted (500 µg) - HDA	Low Dose (250 µg) - LD	Low Dose Adjuvanted (250 µg) - LDA	Placebo saline (Placebo-S)	Placebo adjuvanted (Placebo-SA)
		N = 10	N = 10	N = 10	N = 10	N = 4	N = 4
AGE (years)	Average	26	26	25	26	31	25
BMI (kg/m2)	Average	22.9	23.9	24.3	23.8	23.2	22.6
Serum IgG	Average	0.044	0.051	0.056	0.051	0.051	0.040
(OD450 nm)	SEM	0.008	0.012	0.005	0.003	0.011	0.004
IFN-g	Average	55.4	13.2	9.4	33.0	13.6	11.3
(pg/ml)	SEM	32.2	2.8	0.0	22.1	4.2	1.9
		Lost to follow-up (N = 0)	Lost to follow-up (N = 0)	Lost to follow-up (N = 0)	Lost to follow-up (N = 0)	Lost to follow-up (N = 0)	Lost to follow-up (N = 0)
		High Dose (500 µg) - HD ANALYSED	High Dose Adjuvanted (500 µg) - HDA ANALYSED	Low Dose (250 µg) - LD ANALYSED	Low Dose Adjuvanted (250 µg) - LDA ANALYSED	Placebo saline (Placebo-S) ANALYSED	Placebo adjuvanted (Placebo-SA) ANALYSED
Safety & Immunogenicity		N = 10	N = 10	N = 10	N = 10	N = 4	N = 4

Fig. 1. Trial profile and baseline demographic and clinical data for enrolled volunteers on Day 1 (pre-vaccination). All treated volunteers were included in the safety and immunogenicity analysis. Average indicates the arithmetic mean of the sample, whilst SEM indicates the Standard Error of the Mean. IgG values correspond to the net Mean OD_{450} signal obtained by ELISA against FLU-v-v, with each volunteer's sample tested in triplicate. IFN-gamma values correspond to the cytokine production determined by ELISA following in vitro stimulation of PBMCs with FLU-v for 24 h, with each volunteer's sample tested in triplicate.

HDA: 2% (1/42), Placebo-S: 0% (0/8) and Placebo-SA: 0% (0/12)], malaise [LD: 8% (2/24), LDA: 5% (2/38), HD: 12% (3/25), HDA: 0% (0/42), Placebo-S: 0% (0/8) and Placebo-SA: 0% (0/12)], myalgia [LD: 0% (0/24), LDA: 8% (3/38), HD: 4% (1/25), HDA: 7% (3/42), Placebo-S: 0% (0/8) and Placebo-SA: 8% (1/12)] and nausea [LD: 4% (1/24), LDA: 5% (2/38), HD: 0% (0/25), HDA: 2% (1/42), Placebo-S: 0% (0/8) and Placebo-SA: 0% (0/12)]. Neither vomiting nor chills were reported by any subject during the trial.

Local tolerability analysis, assessed by itching, pain and tenderness as reported by the subject, indicated that the majority of reactions occurred 5 min post-vaccination with most reactions reported as mild and none reported as severe.

There were no clinically significant findings in the individual or mean laboratory measurements, vital signs or ECG measurements amongst the experimental groups.

3.2. Immunogenicity: antibody and cellular

Basic immunogenicity analysis of the FLU-v formulation was a defined endpoint of this study, but this being a first-in-man study, such analysis was subservient to the determination of the vaccine's safety and tolerability. Since a wide range of haematological and biochemical tests were used for this purpose, regulatory bodies agreed to immune responses to the vaccine being determined against the whole formulation, rather than to its individual components.

Pre-vaccination (Day 1) anti-FLU-v IgG Mean OD_{450} readings amongst the experimental groups ranged from 0.040 to 0.056 and were not found to be significantly different (Fig. 1). These Mean OD_{450} values clearly show that volunteers had no pre-existing Ig responses to the Influenza conserved regions found in FLU-v.

No volunteers in the LD, HD or HDA groups developed any detectable anti-FLU-v IgG response at any time post vaccination and, as expected, the same outcome was observed in both placebo groups (Fig. 2). In the LDA group an anti-FLU-v IgG response was detectable on days 14 and 21 post-vaccination (Mean-$OD_{450} \pm$ SEM; 0.263 ± 0.199 and 0.163 ± 0.094, respectively, vs 0.051 ± 0.003 at pre-vaccination). This positive signal, however, resulted from only 30% (3/10) volunteers in the LDA group developing a FLU-v specific antibody response. FLU-v's inability to induce

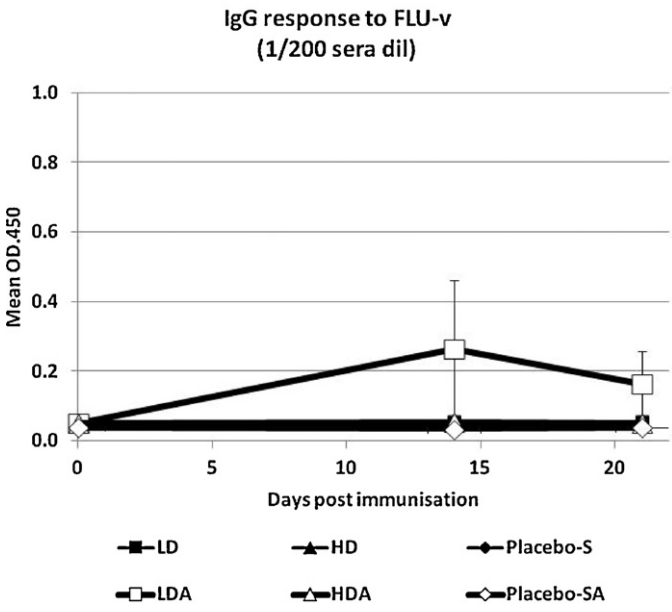


Fig. 2. FLU-v specific antibody response post-vaccination. Analysis includes all volunteers ($N = 48$) and represents the Mean OD_{450} of each experimental group. Error bars represent the Standard Error of the Mean and are provided only for the LDA and the Placebo-SA groups. Only 30% (3/10) of volunteers in the LDA group developed a FLU-v specific IgG response post vaccination. No volunteers in either the remaining experimental groups ($N = 10$ each) or placebo groups ($N = 4$ each) developed FLU-v specific IgG responses.

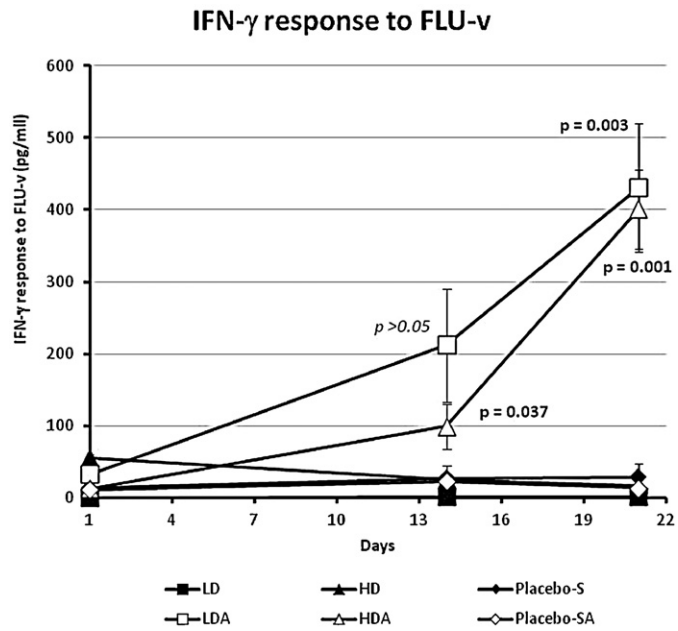


Fig. 3. FLU-v specific cellular immune response post-vaccination. Analysis includes all volunteers ($N = 48$) and represents the in vitro IFN-γ response (pg/ml) to FLU-v of each experimental group. Error bars represent the Standard Error of the Mean and are provided only for the LDA, HDA, Placebo-S and Placebo-SA groups. Statistical significance is represented by the p value achieved in a one-way Mann–Whitney analysis. By Day 21 post-vaccination FLU-v specific IFN-γ responses were detected in 80% (8/10) of volunteers in the LDA group and in 100% (10/10) of volunteers in the HDA group. No volunteers in the LD or HD groups ($N = 10$ each) nor in any of the Placebo group ($N = 4$ each) developed FLU-v specific IFN-γ responses.

strong antibody responses is consistent with our observations in mice immunised with these peptides [9].

Cellular immunity was assessed as IFN-γ production following in vitro challenge with an equimolar preparation of all four peptides in the FLU-v formulation. Pre-vaccination (Day 1) FLU-v specific IFN-γ responses (Mean ± SEM) ranged from 9.4 ± 0.0 to 55.4 ± 32.2 pg/ml (Fig. 1). The higher pre-vaccination FLU-v IFN-γ responses were found in the HD and LDA groups, each of which included a single individual with a pre-vaccination IFN-γ response to FLU-v 2-fold above their non-specific IFN-γ response (i.e. to media). The presence of only two individuals (4.1%) in the whole test population with pre-existing FLU-v-specific IFN-γ responses clearly shows that sustained historical exposure to Influenza does not lead to a strong natural cellular response to FLU-v antigens.

No significant differences were observed amongst groups in their IFN-γ response to a non-specific control (i.e. media) at any time during the study (range 9.4 ± 0.0 to 32.2 ± 16.9). Also, and for the duration of the study, all groups elicited dose-dependent IFN-γ responses to ConA that were 8-fold above (i.e. >280 pg/ml) the maximum IFN-γ response to the non-specific control observed during the study (32.2 ± 16.9 pg/ml).

In the LD, HD and in both placebo groups no volunteer developed a FLU-v specific IFN-γ response post-vaccination (Fig. 3). In contrast, by Day 21 post-vaccination, 80% of volunteers (8/10) in the LDA group had developed FLU-v specific IFN-γ responses 2-fold higher their pre-vaccination response. In the HDA group the percentage of reactive volunteers was higher at 100% (10/10). Despite the apparently higher FLU-v specific IFN-γ response (pg/ml) of the LDA group compared to the HDA group on Day 14 post-vaccination (Fig. 3; 212 ± 78 vs 99 ± 31), a dose-dependent response in favour of HDA is still maintained as the percentage of FLU-v reactive volunteers in the HDA group (80%, 8/10) is higher than in the LDA group (60%, 6/10). This conclusion is sustained by the difference in the response vs placebo on Day 14 being significantly different for the HDA group (99 ± 31 vs 22 ± 9; $p = 0.037$), but not for the LDA group (212 ± 78 vs 26 ± 17; $p > 0.05$).

4. Conclusions

This is a first-in-man evaluation of a synthetically manufactured vaccine (FLU-v) incorporating conserved immunogenic regions identified in NP, M1 and M2 proteins of Influenza.

Being an exploratory study, volunteers received two different non-optimised vaccine doses (250 or 500 μg), delivered in a single vaccination either alone or with an adjuvant.

Vaccination with FLU-v did not cause major AEs. The vast majority were mild and the most common of them, pain at the site of injection, was consistent with the presence of the adjuvant in the formulation, but not with an increase in the dose of the vaccine. These results suggest that FLU-v is safe and well tolerated.

There is an extensive body of literature showing that the use of ISA-51 in clinical trials is generally associated with the range of mild and localised AEs described here (exemplified in [15–20]). Two studies have reported significant induration and even sterile abscesses associated with ISA-51 containing preparations [21,22], but in both cases the high dose of antigen (>1 mg per dose) and/or the specific nature of the antigen most likely played a key role in SAE induction.

The numbers involved in this study are, obviously, too small to determine potential low frequency AEs for FLU-v. Nonetheless, due to its method of manufacture (i.e. Fmoc chemistry), it is possible that FLU-v may have lower intrinsic safety and tolerability risks than traditional Influenza vaccines. Most of these vaccines are manufactured in eggs and contamination by *Campylobacter* and *Salmonella*, which can colonise chickens and contaminate eggs, as well as egg allergy amongst the population are clear risk factors [6]. Also, Influenza vaccination campaigns have occasionally (e.g. 1976) resulted in the increased incidence of Guillain–Barré syndrome, with molecular mimicry to egg-produced sialylated HA being a potential cause [23].

Regarding the immunogenicity of FLU-v, one single vaccination with the adjuvanted formulation induced a dose-dependent FLU-v-specific IFN-γ response in >80% of volunteers in the absence of a significant antibody response. Non-adjuvanted formulations induced neither antibody nor cellular responses, which contrasts with the results of some of our previous animal studies.

No strong pre-vaccination cellular response to the vaccine was observed despite sustained historical exposure to the virus and the fact that the FLU-v antigens are presented by HLA molecules in Influenza infected human cells [9]. These observations suggest that most of the T cell epitopes contained in FLU-v are sub-dominant and natural infection does not induce high frequency cellular responses. Immunodominance, however, is not a pre-requisite for vaccine efficacy. We [9] and others [24,25] have shown that subdominant epitopes are effective vaccine targets for inducing protection against respiratory viral infections. We thus believe the observed post-vaccination FLU-v specific cellular response will be clinically relevant.

The above described immunogenicity profile is uncommon amongst vaccines, and more so amongst Influenza vaccines. Epidemic and pandemic Influenza vaccines currently licensed or in advanced clinical development (reviewed in [26]) elicit neutralising antibody responses to HA/NA, but only limited cellular responses. Only a handful of Influenza vaccines aimed at eliciting T cell responses have been clinically tested. However these either face a difficult regulatory path to market authorisation due to the nature of the vector used (recombinant Modified Vaccinia virus Ankara or MVA) [27] or still rely primarily in the induction of an antibody response against the HA component [15]. Achievement of high antibody titres to HA/NA is hence generally considered as the key marker of effective protection [28], even though several reports [29–32] clearly show that in the elderly, in whom current vaccine efficacy is as low as 35% [5,33], IFN-γ responses to the vaccine are better correlates of vaccine efficacy than antibody titres alone.

In summary, the results of this trial are encouraging as FLU-v has been shown to be well tolerated and immunogenic. Much work remains to be done before the safety and efficacy profile of FLU-v is fully established and our next step will be to test the efficacy of this formulation in a Phase II live Influenza challenge. Nonetheless, the potential applications of a vaccine that stimulates cellular immunity against conserved regions in Influenza is clear and extends beyond the intended use of FLU-v as a stand-alone vaccine. Combined with HA/NA-based epidemic and pandemic vaccines FLU-v could increase their prophylactic efficacy by stimulating anti-viral cellular immune mechanisms not currently exploited by those vaccines. Similarly, in pandemic scenarios, and consistent with the universal view that T cells control and mitigate infection and illness [6,28,34,35], FLU-v could be used to control illness in individuals until a strain specific HA/NA-based vaccine becomes available.

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